Gastric tissue classification

Decomposition-aware stain augmentation

Seventh-semester progress evaluation

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Problem and objectives

Classify eight tissue types from histopathology patches.

Investigate whether decomposition diagnostics can guide stain augmentation.

Evaluate fairly using matched controls and audited data.

Dataset and tissue classes

HMU-GC-HE-30K: 31,096 PNG patches, 224 x 224 pixels.

3,887 images in each of eight classes.

ADI, DEB, LYM, MUC, MUS, NOR, STR and TUM.

Clinical and biomarker tables support dashboard analytics only.

Implemented application

Python and PyTorch for preprocessing and classification.

FastAPI and Uvicorn serve the application.

HTML, CSS and JavaScript provide the dashboard.

SQLite preserves prediction probabilities, confidence and timestamps.

CNN, BiLSTM and attention

Stage	Implemented operation
Image	224 x 224 RGB; sixteen 56 x 56 subpatches
CNN	32/64/128 channels; 128 features per patch
BiLSTM	Two layers; 128 hidden units per direction
Attention	Weighted aggregation of 256-dimensional outputs
Head	256 to 128 to 8 logits; dropout 0.2

Research basis

Prior work	Connection to our study
Lou et al., 2025	Gastric dataset and classifier validation
Stain SAN, 2024	Training-derived stain distribution
DDCA, 2023	Admissibility checking of augmented styles

Candidate distinction: source-decomposition diagnostics control strength.

Experimental augmentation controller

Estimate stain matrix W and concentrations H.

Measure reconstruction error and log condition number.

Scale risk using training-only percentile bounds.

Use lower augmentation strength for higher diagnostic risk.

Preserve the original residual; retain failed-extraction images.

Preprocessing stability results

Noise sigma	Fixed gain	Matched uniform	Adaptive
0.5 RGB units	1.7247	1.4249	1.3049
2.0 RGB units	1.4716	1.2532	1.1646

80 new validation images; lower gain means less noise amplification.

Image-bootstrap intervals included no improvement.

Classifier pilot protocol

512 training images and 128 validation images.

Six augmentation arms, one shared initialization and seed 42.

Three epochs per arm; batch size 16.

Best checkpoints reloaded before validation.

RTX 3050 Laptop GPU, 4 GB; CUDA training verified.

Three-epoch pilot validation

Method	Accuracy (%)	Macro-F1
Standard	33.59	0.2438
Original SAN	32.81	0.2177
Residual fixed	30.47	0.2323
Uniform weaker	33.59	0.2802
Shuffled	35.94	0.2768
Adaptive	35.16	0.2651

Adaptive macro-F1 did not beat the weaker or shuffled controls.

Duplicate audit and limitations

932 exact-file duplicate groups across the dataset.

334 groups crossed historical splits; 165 spanned train and test.

Pilot selection quarantined exact-duplicate identities.

A later provisional manifest retains 29,910 unique-file images.

Patient and slide linkage remains unverified.

Progress and current campaign

Completed: application, controller, numerical checks and duplicate audit.

Completed: four augmentation arms, three matched seeds each.

Adaptive mean macro-F1: 0.6836; fixed: 0.6822.

Mean tumor recall: adaptive 75.56%; standard 79.97%.

Request patient mapping and reserve final test confirmation.

Completed three-seed validation

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Arm	Accuracy mean	Macro-F1 mean ± SD	TUM recall
Standard	67.70%	0.6776 ± 0.0106	79.97%
Residual fixed	68.30%	0.6822 ± 0.0141	74.18%
Uniform weaker	68.22%	0.6804 ± 0.0061	75.56%
Adaptive	68.41%	0.6836 ± 0.0047	75.56%

Adaptive − fixed paired F1: +0.0227, −0.0101, −0.0084. No established superiority.

Same 3,059 validation images across seeds; patient independence unverified.

Measured throughput and restartable training

Setting	Observed result
Cached augmentation, batch 64, four workers	238.7 images/second
Batch 96	Rejected for insufficient VRAM headroom
Exact deterministic cache checks	64 / 64 pixel-identical
Cache extraction failures	2 images retained with unchanged fallback

Main CPU threads: 2. Each loader worker: 1. Mixed precision uses FP16.

Resumable state saves every 120 seconds and each epoch. The 9 AM snapshot does not stop training.

Demonstration and conclusion

Upload a tissue patch and show all class probabilities.

Identify the active inference source.

Show SQLite prediction history.

The application works; the research benefit remains under investigation.